Hypergamy Is Solved

Feasibility, Deficit, and Equilibrium in Status-Ordered Matching Markets

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Abstract

Hypergamy, the tendency to form unions with a partner of higher status, is usually argued about as a behavioural claim. We instead model the marriage market as a coupling problem between Borel probability measures on Polish type spaces and show that, absent taste heterogeneity, every aggregate question about hypergamy reduces to a question about two marginal distributions. We prove: (i) universal hypergamy is feasible if and only if one status distribution first-order stochastically dominates the other (Strassen); (ii) otherwise the minimal unmatched mass is the *Hypergamy Deficit* $\mathsf{D} = \sup_t (G_{\mathfrak{W}} - G_{\mathfrak{M}})^+$ (a continuum Hall marriage theorem); (iii) no behavioural strategy changes any of this (*Cope Invariance*), and status gains are zero-sum in rank (*Conservation of Rank*); (iv) under status-ranked preferences the unique Gale–Shapley stable matching is positive assortative, and under Becker–Shapley–Shubik transferable utility it is the comonotone optimal transport plan, with the Choo–Siow entropic relaxation tending to random matching as taste shocks grow. Height is our worked example: for equal variances the deficit for a required height gap has a closed form that vanishes up to the mean difference of about 14 cm, and the height preference online daters reveal is relative, which the market can satisfy. We chose height because it is measurable, not because it matters most, and review what else people weight: kindness and intelligence in stated long-term preferences, the relationship itself in predicting relationship quality, and looks in stated short-term preferences. With several traits, checklist requirements shrink the eligible pool multiplicatively when traits are independent, trade-offs reduce to a scalar index, and a required rank gap δ leaves exactly δ of one side unmatched. We model partner history as a comparison effect, which applies to both sexes, and show that, for an average new partner, one ex remembered a standard deviation better than they were costs more than four ordinary ones.

1 The Marriage Market as a Measure Space

Definition 1.1 (Market)**.** A *marriage market* is a tuple $(\mathcal{X}, \mathcal{Y}, \mu, \nu, s)$ where $\mathcal{X}$ and $\mathcal{Y}$ are Polish spaces of agent types for the two sides $\mathfrak{M}$ and $\mathfrak{W}$, $\mu \in \mathcal{P}(\mathcal{X})$ and $\nu \in \mathcal{P}(\mathcal{Y})$ are Borel probability measures describing the populations, and $s : \mathcal{X} \sqcup \mathcal{Y} \to \mathbb{R}$ is a measurable *status functional* (height, income, education, follower count, or any scalar index thereof). We write $\mu_s = s_{\#}\mu$ and $\nu_s = s_{\#}\nu$ for the pushforward status distributions, with CDFs $F_{\mathfrak{M}}, F_{\mathfrak{W}}$ and survival functions $G_{\mathfrak{M}}(t) = \mu(s \geq t)$, $G_{\mathfrak{W}}(t) = \nu(s \geq t)$.

Definition 1.2 (Matching)**.** A *matching* is a coupling $\pi \in \Pi(\mu, \nu)$, i.e. a Borel probability measure on $\mathcal{X} \times \mathcal{Y}$ with marginals μ and ν. A *partial matching* is a sub-coupling: a measure π with $(p_1)_{\#}\pi \leq \mu$ and $(p_2)_{\#}\pi \leq \nu$. Its *mass* $|\pi| = \pi(\mathcal{X} \times \mathcal{Y})$ is the amount of each side that is matched.

Definition 1.3 (Hypergamy)**.** A partial matching π is *hypergamous for* $\mathfrak{W}$ if $\pi(\{s(x) \geq s(y)\}) = |\pi|$, and *strictly hypergamous* if the inequality is strict π-a.e. The *hypergamy index* is $\mathsf{H}(\pi) = \pi(\{s(x) > s(y)\})$ and the *hypogamy index* is $\mathsf{H}^{\downarrow}(\pi) = \pi(\{s(x) < s(y)\})$.

Remark 1.4. Every definition is symmetric under exchanging $(\mathcal{X}, \mu)$ with $(\mathcal{Y}, \nu)$. Conventional usage applies hypergamy to $\mathfrak{W}$, and we follow that convention for notation only. No result depends on which side is labelled which, and the reader is free to swap the letters if it makes them feel better.

2 Who Can Marry Up? Feasibility and the Deficit

Theorem 2.1 (Hypergamy Feasibility)**.** *A hypergamous matching $\pi \in \Pi(\mu, \nu)$ exists if and only if μ_s first-order stochastically dominates ν_s:*

$$G_{\mathfrak{M}}(t) \geq G_{\mathfrak{W}}(t) \qquad \text{for all } t \in \mathbb{R}. \tag{1}$$

Proof. ($\Rightarrow$) If π is hypergamous then $\{s(y) \geq t\} \subseteq \{s(x) \geq t\}$ up to a π-null set, so $G_{\mathfrak{W}}(t) = \pi(s(y) \geq t) \leq \pi(s(x) \geq t) = G_{\mathfrak{M}}(t)$. ($\Leftarrow$) With generalised inverses $F^{-1}(u) = \inf\{t : F(t) \geq u\}$, condition (1) gives $F_{\mathfrak{M}}^{-1}(u) \geq F_{\mathfrak{W}}^{-1}(u)$ for all u, so for $U \sim \mathrm{Unif}(0,1)$ the pair $(F_{\mathfrak{M}}^{-1}(U), F_{\mathfrak{W}}^{-1}(U))$ couples (μ_s, ν_s) on $\{a \geq b\}$. Lift to $\mathcal{X} \times \mathcal{Y}$ by disintegrating μ, ν along the fibres of s (possible since the spaces are Polish). This is the one-dimensional case of Strassen's theorem [36]. □

When (1) fails, someone has to marry down or stay single. We compute exactly how many. Allow general masses $\mu(\mathcal{X}) = m$, $\nu(\mathcal{Y}) = w$ and let $\mathcal{C} = \{(x, y) : s(x) \geq s(y)\}$.

Theorem 2.2 (Hypergamy Deficit Formula)**.** *The maximal mass of a partial matching supported on $\mathcal{C}$ is*

$$\max_{\pi(\mathcal{C}^c)=0} |\pi| = w - \mathsf{D}, \qquad \mathsf{D} := \sup_{t \in \mathbb{R}} \big(\nu(s \geq t) - \mu(s \geq t)\big)^+. \tag{2}$$

D *is the minimal mass of $\mathfrak{W}$ left unmatched if every union is hypergamous.*

Proof. Upper bound. Agents in $\mathfrak{W}$ with $s(y) \geq t$ are compatible only with $\{x : s(x) \geq t\}$, so at least $\nu(s \geq t) - \mu(s \geq t)$ of them are unmatched; take the supremum. *Achievability.* Leave the top D of $\mathfrak{W}$ by status unmatched, splitting an atom if necessary, and use only the top $w - \mathsf{D} \leq m$ of $\mathfrak{M}$. These women and men have survival functions $(G_{\mathfrak{W}} - \mathsf{D})^+$ and $\min(G_{\mathfrak{M}}, w - \mathsf{D})$, and the second dominates the first because $G_{\mathfrak{W}} - G_{\mathfrak{M}} \leq \mathsf{D}$ and $G_{\mathfrak{W}} \leq w$, so Theorem 2.1, rescaled to mass $w - \mathsf{D}$, matches them hypergamously. For finitely supported measures this is the deficiency form of Hall's theorem: compatibility is *nested*, so Hall's condition need only be checked on upper sets. □

Corollary 2.3. $\mathsf{D} = 0$ *iff* (1) *holds (when $m = w$), and* $\mathsf{D} \geq (w - m)^+$. *A skewed sex ratio is a lower bound on the deficit; everything above it is caused by the shape of the distributions.*

Example 2.4. Let $\mu_s = \mathrm{Unif}[0,1]$ and $\nu_s = \mathrm{Beta}(2,1)$, so $G_{\mathfrak{M}}(t) = 1 - t$ and $G_{\mathfrak{W}}(t) = 1 - t^2$. Then $\mathsf{D} = \sup_t (t - t^2) = \frac{1}{4}$ at $t^* = \frac{1}{2}$. With a perfectly balanced sex ratio, one quarter of $\mathfrak{W}$ still cannot be matched hypergamously, and the bottleneck sits exactly at the median of $\mathfrak{M}$.

3 Application: The Six-Foot Rule

Height is the cleanest test case: scalar, observable, approximately Gaussian within each sex, and the subject of strongly held opinions. It is chosen for convenience, not because it matters most, and Section 10 reviews what else does. Take $s =$ height in cm with $\mu_s = N(m_{\mathfrak{M}}, \sigma^2_{\mathfrak{M}})$ and $\nu_s = N(m_{\mathfrak{W}}, \sigma^2_{\mathfrak{W}})$. We calibrate to US adult means from NHANES 2015–2018 [19], $m_{\mathfrak{M}} \approx 175.3$ and $m_{\mathfrak{W}} \approx 161.3$, write $\Delta m \approx 14.0$, and take $\sigma_{\mathfrak{M}} = 7.6$ and $\sigma_{\mathfrak{W}} = 7.1$. The report gives percentiles rather than standard deviations, and its percentiles imply 7.4–7.5 and 6.9–7.1.

Proposition 3.1 (The -37.5 cm Theorem)**.** *For Gaussian marginals with $\sigma_{\mathfrak{M}} \neq \sigma_{\mathfrak{W}}$, the survival functions cross exactly once, at*

$$t_\times = \frac{\sigma_{\mathfrak{M}} m_{\mathfrak{W}} - \sigma_{\mathfrak{W}} m_{\mathfrak{M}}}{\sigma_{\mathfrak{M}} - \sigma_{\mathfrak{W}}} \approx -37.5\,\text{cm}. \tag{3}$$

Hence (1) fails only on $(-\infty, t_\times)$ and $\mathsf{D}(0) < 10^{-150}$.

Proof. $G_{\mathfrak{M}}(t) = G_{\mathfrak{W}}(t)$ iff $(t - m_{\mathfrak{M}})/\sigma_{\mathfrak{M}} = (t - m_{\mathfrak{W}})/\sigma_{\mathfrak{W}}$, a linear equation with unique root $t_\times$. Above it, $\mathfrak{M}$ has the smaller standardised score, so $G_{\mathfrak{M}} > G_{\mathfrak{W}}$. Below it the gap is at most Φ of a standardised value near -28. □

Remark 3.2. "He must be taller than me" is therefore feasible for every human of positive height. The only obstruction occurs below -37.5 cm, a region we were unable to sample. Other readings of the report's percentiles move the crossing anywhere from -6 to -108 cm, which we did not sample either.

Now require a gap. A union satisfies the *δ-rule* if $s(x) \geq s(y) + \delta$. Shifting ν_s by δ reduces this to Theorem 2.2, so $\mathsf{D}(\delta) = \sup_t (G_{\mathfrak{W}}(t) - G_{\mathfrak{M}}(t + \delta))^+$.

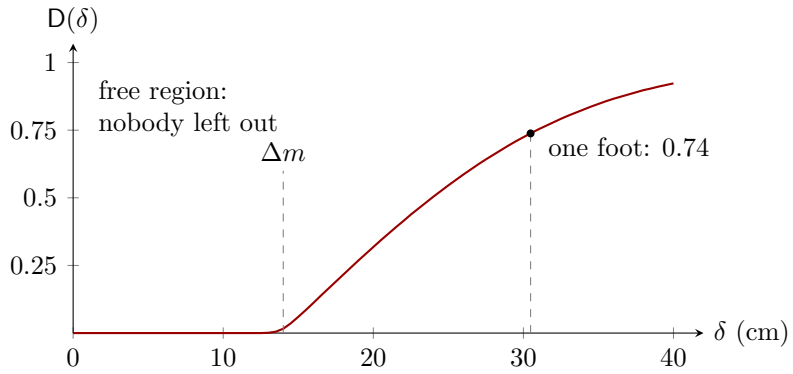


Figure 1: Minimal fraction of $\mathfrak{W}$ left unmatched as a function of the required height gap, US calibration with unequal variances. Essentially zero until about 13 cm, then roughly 5% per centimetre.

Theorem 3.3 (Closed-Form Height Deficit)**.** *If $\sigma_{\mathfrak{M}} = \sigma_{\mathfrak{W}} = \sigma$, then*

$$\mathsf{D}(\delta) = \left(2\,\Phi\left(\frac{\delta - \Delta m}{2\sigma}\right) - 1\right)^+. \tag{4}$$

D vanishes iff $\delta \leq \Delta m$, and is concave increasing beyond it with initial slope $(\sigma\sqrt{2\pi})^{-1}$, or 5.4% per cm *at the pooled $\sigma = 7.35$* cm.

Proof. With $z = (t - m_{\mathfrak{W}})/\sigma$ and $c = (\delta - \Delta m)/\sigma$, the gap is $\Phi(z + c) - \Phi(z)$, the Gaussian mass of an interval of length c. For $c > 0$ it is maximised by centring the interval at the mode, $z = -c/2$, giving $2\Phi(c/2) - 1$. Its δ-derivative $\varphi(c/2)/\sigma$ is decreasing in c. □

Theorem 3.4 (The Six-Foot Theorem)**.** *Suppose a fraction α of $\mathfrak{W}$ requires $s(x) \geq \tau$, and everyone else imposes any constraint whatsoever. Then at least $(\alpha - G_{\mathfrak{M}}(\tau))^+$ of $\mathfrak{W}$ is unmatched in every feasible matching. For $\tau = 182.9$ cm (six feet), $G_{\mathfrak{M}}(\tau) \approx 0.16$.*

Proof. The constrained set A has neighbourhood inside $\{s \geq \tau\}$, of mass $G_{\mathfrak{M}}(\tau)$. Apply the Hall bound from Theorem 2.2. □

Remark 3.5. Theorem 3.4 holds regardless of personality, effort, or app. It is also generous: it assumes every six-footer is available and interested.

4 The Blackpill Theorems

Lemma 4.1 (Conservation of Status)**.** *If $s \in L^1(\mu) \cap L^1(\nu)$, then for every $\pi \in \Pi(\mu, \nu)$, $\int (s(x) - s(y))\, d\pi = \int t\, d\mu_s - \int t\, d\nu_s$. The average status gap across all couples does not depend on who marries whom.*

Proof. Integrate against π and use its marginals. □

Definition 4.2 (Strategy)**.** A *strategy* is any map $\mathfrak{S} : \Pi(\mu, \nu) \to \Pi(\mu, \nu)$: every intervention that changes who matches with whom without changing the population. This includes openers, dating apps, "just be confident," and every other object the literature calls *cope*.

Theorem 4.3 (Cope Invariance)**.** *If $\Psi : \Pi(\mu, \nu) \to \mathbb{R}$ depends on π only through its marginals, then $\Psi \circ \mathfrak{S} = \Psi$ for every strategy $\mathfrak{S}$. In particular the mean status gap, the feasibility of universal hypergamy, the deficit D, and the equilibrium index H^* of (8) are invariant under every strategy.*

Proof. Strategies preserve marginals, and each listed quantity is a functional of (μ_s, ν_s) alone. □

Corollary 4.4 (It's Over)**.** *If $\mu_s = \nu_s$, no strictly hypergamous matching exists and no strategy produces one: by Lemma 4.1 applied to the bounded status $\arctan s$, the integrand $\arctan s(x) - \arctan s(y)$ would be positive π-a.e. with zero integral.*

The only escape is to change the marginals, replacing s by $s + h$ with $h \geq 0$ on a subset of agents. We call this *maxxing*.

Theorem 4.5 (Conservation of Rank)**.** *Let $F_{\mathfrak{M}}, F_{\mathfrak{W}}$ be continuous and let the market clear at the comonotone coupling. If $U = F_{\mathfrak{M}}(s(x))$ is an agent's own quantile and $V = F_{\mathfrak{W}}(s(y))$ their partner's, then $V = U$ a.s. and $U \sim \mathrm{Unif}(0, 1)$. Hence for any perturbation $s \mapsto s + h$ that keeps $F_{\mathfrak{M}}$ and $F_{\mathfrak{W}}$ continuous, maxxing included,*

$$\mathbb{E}[V] = \tfrac{1}{2}, \qquad \mathrm{Var}(V) = \tfrac{1}{12} \tag{5}$$

before and after. Maxxing by a subpopulation raises its partners' quantiles by exactly the amount it lowers everyone else's.

Proof. Under the comonotone coupling $s(y) = F_{\mathfrak{W}}^{-1}(F_{\mathfrak{M}}(s(x)))$, so $V = U$. The probability integral transform gives $U \sim \mathrm{Unif}(0, 1)$ for any continuous $F_{\mathfrak{M}}$, perturbed or not. □

Remark (The Blackpill, Stated Precisely). Theorems 4.3 and 4.5 are the rigorous content of "it's over": aggregate outcomes are pinned by the marginals, strategies act within a fibre of fixed marginals, and status gains are transfers of rank, not creation of it. But both are statements about the *population*. For an individual at quantile u the comonotone map is strictly increasing, so raising your own status raises your partner's quantile one-for-one, and for $\theta > 0$ (Section 7) outcomes are random rather than determined.

It is over for the population. It is not over for you.

5 Everyone Wants Up, Almost Everyone Goes Down

Take finite sides $|\mathfrak{M}| = |\mathfrak{W}| = n$ with distinct statuses and *status-ranked* preferences: every agent strictly prefers a higher-status partner, and any partner to none. A matching is *stable* if no pair would both rather be with each other than with their assigned partners.

Theorem 5.1 (Uniqueness of the Assortative Equilibrium)**.** *Under status-ranked preferences the unique stable matching pairs the k-th highest of $\mathfrak{M}$ with the k-th highest of $\mathfrak{W}$ for every k, and Gale–Shapley deferred acceptance [20] returns it whichever side proposes.*

Proof. Induct on n. If the two top agents are not matched to each other, they form a blocking pair. Remove them; the reduced market has the same structure. □

Corollary 5.2 (Everyone Wants Up, Almost Everyone Goes Down)**.** *In Example 2.4 the assortative matching is the quantile map $T = F_{\mathfrak{M}}^{-1} \circ F_{\mathfrak{W}}$, $T(b) = b^2 < b$ on $(0,1)$, and no pair blocks it: a man at quantile u prefers a woman at quantile v only if $v > u$, and she prefers him only if $u > v$. Hence $\mathsf{H} = 0$ and $\mathsf{H}^{\downarrow} = 1$: every member of $\mathfrak{W}$ prefers to marry up, the market is stable, and almost every member of $\mathfrak{W}$ marries down, by an average of $\int_0^1 (b - b^2)\, 2b\, db = \frac{1}{6}$.*

6 Transferable Utility and Optimal Transport

Following Becker [7] and Shapley–Shubik [33], a couple with statuses a and b generates surplus $f(a,b)$, which partners may split. The equilibrium solves the Monge–Kantorovich problem

$$\sup_{\pi \in \Pi(\mu,\nu)} \int f(s(x), s(y))\, d\pi = \inf_{u(x)+v(y) \geq f(s(x),s(y))} \left\{ \int u\, d\mu + \int v\, d\nu \right\}, \tag{6}$$

where equality is Kantorovich duality and (u, v) are the equilibrium utilities [38].

Theorem 6.1 (Becker via Cyclical Monotonicity)**.** *If f is strictly supermodular ($\partial_{ab} f > 0$), every optimal plan has status projection equal to the comonotone coupling $\gamma^* = (F_{\mathfrak{M}}^{-1}, F_{\mathfrak{W}}^{-1})_{\#}\,\mathrm{Unif}(0,1)$.*

Proof. Optimal plans are cyclically monotone. If $a_1 < a_2$ and $b_1 > b_2$ were both in the support, then

$$f(a_2, b_1) + f(a_1, b_2) - f(a_1, b_1) - f(a_2, b_2) = \int_{a_1}^{a_2} \int_{b_2}^{b_1} \partial_{ab} f\, d\beta\, d\alpha > 0, \tag{7}$$

so swapping partners strictly improves total surplus, a contradiction. □

So bargaining or not, the realised hypergamy index is the same:

$$\mathsf{H}^* := \mathsf{H}(\gamma^*) = \mathrm{Leb}\{u \in (0,1) : F_{\mathfrak{M}}^{-1}(u) > F_{\mathfrak{W}}^{-1}(u)\}. \tag{8}$$

7 Taste Heterogeneity: The Entropic Relaxation

Real agents are not perfectly status-ranked. Following Choo and Siow [11] and Galichon and Salanié [21], add i.i.d. Gumbel taste shocks of scale $\theta > 0$. With finitely many types, write f_{ij} for the surplus of a type-i man with a type-j woman. Equilibrium couples π_{ij} and singles π_{i0}, π_{0j} satisfy

$$\pi_{ij} = \sqrt{\pi_{i0}\pi_{0j}}\, e^{f_{ij}/(2\theta)}, \tag{9}$$

the first-order conditions of the entropy-regularised problem

$$\max_{\pi \geq 0} \Big\{ \sum_{i,j} f_{ij}\pi_{ij} - \theta \operatorname{Ent}(\pi) \Big\}, \qquad \operatorname{Ent}(\pi) = 2\sum_{i,j} \pi_{ij} \log \pi_{ij} + \sum_i \pi_{i0} \log \pi_{i0} + \sum_j \pi_{0j} \log \pi_{0j}, \tag{10}$$

subject to the population constraints. Writing $a_i = \sqrt{\pi_{i0}}$, $b_j = \sqrt{\pi_{0j}}$, $K_{ij} = e^{f_{ij}/(2\theta)}$, these become $a_i^2 + a_i(Kb)_i = \mu_i$ and $b_j^2 + b_j(K^\top a)_j = \nu_j$, solved alternately in closed form by a generalised iterative proportional fitting procedure.

Theorem 7.1 (Vanishing and Exploding Taste)**.** *Let the market be balanced and f strictly supermodular. If every $f_{ij} > 0$, then as $\theta \downarrow 0$ the equilibrium converges to an optimal plan of (6) and $\mathsf{H} \to \mathsf{H}^*$. In the variant without singles this holds for any sign of f_{ij}, and as $\theta \to \infty$ the equilibrium converges to $\mu \otimes \nu$ and $\mathsf{H} \to \mathbb{P}(S_{\mathfrak{M}} > S_{\mathfrak{W}})$ for independent $S_{\mathfrak{M}} \sim \mu_s$, $S_{\mathfrak{W}} \sim \nu_s$.*

Sketch. Γ-convergence of the regularised objectives on a compact feasible set. If every $f_{ij} > 0$, a man and a woman left single can be matched for a gain, so limit plans have no singles, and Theorem 6.1 fixes their status projection, on which H depends; for $\theta \to \infty$ the entropy term dominates and its unique minimiser over $\Pi(\mu, \nu)$ is the product measure. □

In Example 2.4 the endpoints are $\mathsf{H}^* = 0$ and $\mathbb{P}(S_{\mathfrak{M}} > S_{\mathfrak{W}}) = \frac{1}{3}$. For height, they are $\mathsf{H}^* = 1$ and $\Phi(\Delta m/\sqrt{\sigma_{\mathfrak{M}}^2 + \sigma_{\mathfrak{W}}^2}) \approx 0.91$. Caring *less* about status produces *more* marrying up in the first case. Preferences and outcomes are different objects.

8 Main Result

Theorem 8.1 (Hypergamy Is Solved)**.** *For any marriage market:*

(a) universal hypergamy is feasible iff $\mu_s \succeq_{\mathrm{FSD}} \nu_s$;

(b) otherwise the minimal unmatched mass is $\mathsf{D} = \sup_t (G_{\mathfrak{W}} - G_{\mathfrak{M}})^+$;

(c) no strategy changes (a) or (b), and for continuous $F_{\mathfrak{M}}, F_{\mathfrak{W}}$ maxxing is zero-sum in rank;

(d) the equilibrium is the comonotone coupling, with index $\mathsf{H}^* = \mathrm{Leb}\{F_{\mathfrak{M}}^{-1} > F_{\mathfrak{W}}^{-1}\}$, under strictly supermodular transferable utility and under status-ranked preferences in the finite markets of Theorem 5.1;

(e) when taste shocks of scale θ are added to the transferable utility of (d), with finitely many types and no singles, the realised index tends to H^* as $\theta \downarrow 0$ and to $\mathbb{P}(S_{\mathfrak{M}} > S_{\mathfrak{W}})$ as $\theta \to \infty$.

Every quantity in (a)–(d) is a functional of (μ_s, ν_s) alone, and (e) adds only θ and the surplus f. No other behavioural input is required.

Proof. Theorems 2.1, 2.2, 4.3, 4.5, 5.1, 6.1, and 7.1. □

Remark 8.2 (Interpretation). When one side's status distribution overtakes the other's, measured hypergamy falls even if no one's preferences change. This matches the reversal of the gender gap in education documented by Esteve et al. [17], who report a global shift from educational hypergamy towards hypogamy.

9 Beyond Height: Multidimensional Status

Height is one coordinate of a longer list. Let each man carry a trait vector $x = (x_1, \dots, x_k)$ (height, face, confidence, charisma, kindness, income, and so on), each coordinate a within-sex z-score, so that μ is now a distribution on $\mathbb{R}^k$. A requirement can combine traits in two ways: as a checklist, where every trait must clear its own bar, or as a trade-off, where strength in one trait offsets weakness in another.

Theorem 9.1 (The Checklist Theorem)**.** *Suppose a fraction α of $\mathfrak{W}$ requires every trait to clear a bar, $x \in A = \{x_1 \geq \tau_1, \dots, x_k \geq \tau_k\}$. Then at least $(\alpha - \mu(A))^+$ of $\mathfrak{W}$ is unmatched in every feasible matching, and for independent traits $\mu(A) = \prod_i \mu(x_i \geq \tau_i)$.*

Proof. The constrained women's neighbourhood lies inside A, of mass $\mu(A)$. Apply the Hall bound from Theorem 2.2, exactly as in Theorem 3.4. □

Example 9.2. In the calibration of Section 3, six feet is $x_1 \geq 1$, so $\mu(x_1 \geq 1) \approx 0.16$. Adding a top-20% face and top-30% confidence leaves $\mu(A) \approx 0.16 \times 0.2 \times 0.3 \approx 0.01$ for independent traits. With a common pairwise correlation ρ between traits (a one-factor Gaussian model), $\mu(A)$ rises to 0.032 at $\rho = 0.3$ and 0.054 at $\rho = 0.5$. For independent traits, bars multiply: every requirement added shrinks the eligible pool by its own pass rate. Correlation softens this without removing it: at $\rho = 0.5$ the face bar keeps 46% of the six-footers, and the confidence bar 73% of those left.

Proposition 9.3 (Compensation)**.** *If instead $\mathfrak{W}$ ranks men by an index $s(x) = \beta \cdot x$, then s is a scalar status functional, and Theorem 8.1 applies to it once s is also defined on $\mathfrak{W}$ on the same scale. For independent standard Gaussian traits, the fraction of the top $\Phi(-c)$ of men by index who fall below c on trait i is*

$$1 - \frac{\Phi_2(-c, -c; r_i)}{\Phi(-c)}, \qquad r_i = \frac{\beta_i}{\|\beta\|}, \tag{11}$$

where $\Phi_2(\cdot, \cdot; r)$ is the standard bivariate normal CDF with correlation r.

Proof. $(x_i, s/\|\beta\|)$ is standard bivariate normal with correlation r_i, and the top $\Phi(-c)$ of men by index is the event $s/\|\beta\| \geq c$. □

Remark 9.4. With equal weights on height, face and confidence ($r_i = 1/\sqrt{3}$) and $c = 1$, take the top 16% of men by index, as many men as are six feet tall: 56% of them are under six feet. Under a checklist six feet is a gate; under a trade-off it is one input among several. The Six-Foot Theorem is a statement about the checklist.

Nothing requires the two sides to be scored on the same traits. Let $\mathfrak{W}$ rank men by an index $s_{\mathfrak{M}}$ and $\mathfrak{M}$ rank women by a different index $s_{\mathfrak{W}}$. Scores on different traits share no unit, so compare positions instead: write $U = F_{\mathfrak{M}}(s_{\mathfrak{M}}(x))$ and $V = F_{\mathfrak{W}}(s_{\mathfrak{W}}(y))$ for each person's quantile within their own side.

Theorem 9.5 (Quantile Hypergamy)**.** *In a balanced market with continuous score distributions, if every member of $\mathfrak{W}$ requires $U \geq V + \delta$, where $0 \leq \delta \leq 1$, the minimal unmatched mass of $\mathfrak{W}$ is exactly δ, whatever traits define $s_{\mathfrak{M}}$ and $s_{\mathfrak{W}}$.*

Proof. By the probability integral transform U and V are both $\mathrm{Unif}(0,1)$. Shifting by δ as in Section 3, Theorem 2.2 gives $\mathsf{D}(\delta) = \sup_{t\in[0,1]}\big((1-t)-(1-t-\delta)^+\big)^+ = \delta$, attained at every $t \in [0, 1-\delta]$. □

Remark 9.6. At $\delta = 0$ the two quantile distributions coincide, so by Corollary 4.4 no strictly hypergamous matching exists. Once status means rank, a market in which everyone marries strictly up is not rare but impossible: by Lemma 4.1, every rank gained is a rank lost elsewhere.

10 What Else Matters

Height is one coordinate of the trait vector of Section 9, which left the weights β unspecified. Here is what the evidence says about the others. In 37 samples from 33 countries, both sexes ranked kindness and intelligence above physical attractiveness and earning capacity; men rated good looks as significantly more important than women did in 34 samples, and women rated good financial prospects as significantly more important than men did in 36 [10]. When designing a long-term partner on a budget, men treat physical attractiveness as a necessity, women status and resources, and both kindness and intelligence [28]. For short-term partners, women, like men, put physical attractiveness first [29]. A review of the evidence on sex drive finds it stronger in men than in women, with no contrary findings [6].

Stated preferences predict little of what happens next. In speed dating, the sex differences in the stated importance of looks and earnings do not appear in whom people actually like, and stated ideals do not predict it [16]. Across 43 longitudinal studies of couples, features of the relationship itself, led by how committed the partner seems, appreciation, sexual satisfaction, how satisfied the partner seems and conflict, predict up to 45% of the variance in relationship quality at the start of a study and 18% at the end, against 21% and 12% for individual differences such as mood and attachment style, and neither individual differences nor the partner's own reports add anything to what a person reports about the relationship [25]. The relationship itself is the best predictor of how it goes.

Money counts for whoever weights it. Among 6,485 users of an online dating service in Boston and San Diego, an income of \$150k to \$200k rather than \$35k to \$50k raised the chance that a woman contacted a man by 8.9 percentage points, and that a man contacted a woman by 3.9, though few users, and especially few women, reported incomes that high [22]. The same users show what height preferences look like in practice. Women were much less likely to contact men shorter than themselves, and a man more than five inches taller, rather than within two inches of her height, raised a woman's chance of contacting him by 7.1 points, about as much as looks in the 81st to 90th percentile rather than the bottom decile (7.9). There was no evidence that women cared about absolute height, and men's preference for women at least 5 foot 3 was small. Height matters, then, as a relative comparison, and the market can meet it: even as a requirement, five inches is below the mean difference of about 14 cm and leaves about one woman in eight thousand unmatched in the calibration of Section 3. It is the absolute bar of Theorem 3.4 that excludes, and these women show no sign of one.

Short-term play adds game, which is a strategy in the sense of Definition 4.2 when it changes who meets whom, and maxxing when it raises the index itself, since confidence and charisma are on the list. Either way Theorems 4.3 and 4.5 bound what it does for the population, and the boxed remark in Section 4 says what it does for an individual. For men, we emphasise the teachings of Jake Abdo [1] as optimal play for short-term dating; optimality is not proved here.

Uncertainty is part of the game. Women told that a man's interest in them was uncertain were more attracted to him than women told that he liked them a lot, and thought about him more [39]. If someone is already committed to pursuing you, playing hard to get raises how much they want you but lowers how much they like you; otherwise it lowers both [12]. A partner's attachment security also predicts attraction [27], which suggests that unavailability at best trades liking for wanting.

Mystery has a shelf life. Learning more about a person can lower liking [30], though more time in live conversation raises liking between strangers [32], and among newlyweds it was disillusionment, a fading of love and affection over the first two years, that marked the couples headed for divorce [23]. Mystery works like the idealisation ι of Section 12, applied to a current partner, and information removes it. The uncertainty finding comes from women only [39]; whether it holds for men is untested there.

Narcissism helps in the short run. Among 224 people, the dark-triad traits of narcissism, psychopathy and Machiavellianism went with short-term but not long-term mating, more strongly in men [26]. Across three studies, including courtship in natural settings, narcissism went with greater short-term mate appeal, through its self-promoting side rather than its lack of warmth, and via physical attractiveness and social boldness [14]; in speed dating it went with greater appeal in both sexes [24]. At first sight narcissists are more popular, and the facets that do most damage in the long run, exploitativeness and entitlement, are the most attractive [4]. What is rewarded, then, is not giving less but the looks and boldness that come with it, and Li and Kenrick read women's short-term preference for looks as a search for good genes [29]. In the sense of Section 4 this is maxxing, so by Theorem 4.5 whatever rank narcissists gain, someone else loses. The appeal does not last: in groups that met weekly, peers who first saw self-enhancers as agreeable and competent rated them negatively by the seventh meeting [31], and Section 12 describes what their partners are left with.

In the notation of Section 9, each side ranks the other by its own index, which Theorem 9.5 allows, with stated weights that change with the horizon: kindness and intelligence in the long run, looks in the short run, and money for whoever is looking for it. The relationship itself is no one's trait. Its nearest counterpart in Section 7 is the taste shock, the part of the surplus that status does not explain, although Choo and Siow attach each shock to a person and a type of partner rather than to a couple. In the variant without singles, Theorem 7.1 says the equilibrium tends to random matching as the shocks grow relative to the status surplus.

11 Partner History

A common claim holds that fewer previous partners make for better relationships, either because the capacity to pair-bond gets used up or because a partner with no history has no one to compare you to. The data support an association, but not one confined to women. Among 2,654 married individuals, Busby, Willoughby and Carroll [9] find that more lifetime partners go with lower sexual quality, poorer communication and lower stability, with no significant difference between men and women. Using longitudinal data on both sexes, Smith and Wolfinger [34] find divorce risk highest with six or more premarital partners and next highest with one or two, while three to five do not differ significantly from none besides the eventual spouse; the link survives controls for religion, values, family relationships, personality and mental health, shows no gender difference, and selection is not ruled out.

Wolfinger [40] reports the lowest divorce rates for women with zero or one premarital partners, an irregular pattern above that, and religiosity as part of the explanation for the low end; that analysis covers only women because the survey lacks full data on men's premarital sex. The neurobiology of pair bonding comes from studies of monogamous rodents [41], and its human version, in which bonding capacity is used up by previous partners, has not been established. We therefore treat partner history as a hypothesis, and treat it symmetrically.

Definition 11.1 (Comparison)**.** A person whose previous partners had qualities $Q_1, \dots, Q_n$, drawn i.i.d. from a distribution F with finite mean, values a new partner of quality q at $q - \lambda(M_n - q)^+$, where $M_n = \max_{i \le n} Q_i$, $M_0 = -\infty$, and $\lambda \ge 0$ measures how much comparison matters. Only an ex who was better than the new partner costs anything.

Proposition 11.2 (Diminishing Comparisons)**.** *The expected regret* $R_n(q) = \mathbb{E}[(M_n - q)^+]$ *is increasing and concave in* n*, with* $R_0 = 0$ *and*

$$R_{n+1}(q) - R_n(q) = \int_q^\infty F(t)^n \big(1 - F(t)\big)\, dt. \tag{12}$$

For standard Gaussian quality and an average new partner ($q = 0$)*,* $R_n = 0,\ 0.40,\ 0.68,\ 0.89,\ 1.17,\ 1.54,\ 1.87$ *at* $n = 0,\ 1,\ 2,\ 3,\ 5,\ 10,\ 20$*.*

Proof. $R_n(q) = \int_q^\infty \mathbb{P}(M_n > t)\, dt = \int_q^\infty \big(1 - F(t)^n\big)\, dt$, which is finite because F has a finite mean. Subtract consecutive terms; the integrand $F^n(1 - F)$ is nonnegative and decreasing in n because $0 \le F \le 1$. □

Remark 11.3. The first partner sets the bar and each later one raises it by less, so the largest single step is from none to one, though later partners still add most of the total: the first supplies 45% of R_3 and 21% of R_{20}. This fits Wolfinger's jump from one premarital partner, usually the husband, to two [40], but not his higher divorce rate at two than at three to nine, nor Smith and Wolfinger [34], whose results suggest a link that grows stronger, not weaker, as partners accumulate. The model cannot tell comparison apart from selection, and nothing in it refers to sex: a man with many previous partners compares too. If partner history enters a status index, Theorem 4.5 applies unchanged: it re-sorts who ranks where, and every rank one person gains, someone else loses.

12 Abusive Partners

Definition 11.1 charges nothing for a bad ex, since only an ex who was better than the new partner costs anything. An abusive ex can still cost something by being remembered as better than they were. "Narcissistic abuse" is not a clinical diagnosis, and the studies below rate a partner's traits through the person they abused. Among 436 people close to someone high in narcissistic traits, mostly current or former partners, Day, Townsend and Grenyer [13] find abuse alongside mutual idealisation and devaluation. Among 75 women who had just left abusive partners, Dutton and Painter [15] find attachment to the former partner predicted by how extreme and intermittent the maltreatment was and by the imbalance of power, both at separation and six months later. Among 1,294 people recruited through online communities about narcissism and abuse, 94% of them women, Arabi [3] finds that a former partner's narcissism predicts symptoms of post-traumatic stress, which include intrusive memories, beyond the frequency of the abuse and any earlier abuse.

Definition 12.1 (Idealisation)**.** A previous partner is *idealised by* $\iota \ge 0$ if they are remembered at quality $Q_i + \iota$. With one idealised partner the comparison point is $M_n^\iota = \max(Q_1 + \iota, Q_2, \dots, Q_n)$, and since the Q_i are i.i.d. it does not matter which partner it was. Write $R_n^\iota(q) = \mathbb{E}[(M_n^\iota - q)^+]$.

Proposition 12.2 (Idealisation Compounds)**.** $R_n^\iota(q)$ *is nondecreasing and convex in* ι*, with* $R_n^0 = R_n$ *and* $R_1^\iota(q) = R_1(q - \iota)$*. For standard Gaussian quality and an average new partner* ($q = 0$)*,* $R_1^\iota = \iota\,\Phi(\iota) + \varphi(\iota)$*. At* $\iota = 1$ *this is* 1.08*, which is* 2.7 *times* R_1 *and exceeds* $R_4 = 1.05$*, the regret from four ordinary previous partners; at* $\iota = 1.5$ *it is* 1.53*, almost* $R_{10} = 1.54$*.*

Proof. Write $M_n^\iota = \max(Q_1 + \iota, M')$ with M' free of ι. It is convex and nondecreasing in ι, and composing with the convex nondecreasing map $x \mapsto (x - q)^+$ preserves both properties; take expectations, which are finite because F has a finite mean. Also $(Q_1 + \iota - q)^+ = (Q_1 - (q - \iota))^+$. For $Z \sim N(0,1)$, since $\varphi'(z) = -z\varphi(z)$, $\mathbb{E}[(Z + \iota)^+] = \int_{-\iota}^\infty (z + \iota)\varphi(z)\, dz = \varphi(\iota) + \iota\,\Phi(\iota)$. □

Partners have diminishing returns (Proposition 11.2) and idealisation has increasing ones, so in this model one idealised ex can outweigh several ordinary ones, and abuse need not act through n at all. The expected cost of comparison is $\lambda R_n^{\iota}(q)$, so if abuse also raises λ, as the attachment findings suggest, it multiplies all of the above. No study measures ι. Nothing in the model refers to sex, although the samples in [13], [15] and [3] are mostly or entirely women.

Remark 12.3 (First Partners). Since it does not matter which partner was idealised, what is special about a first partner lies outside the model. While that relationship lasts, $M_0 = -\infty$: there is no one to compare them with. Abuse may also lower the floor. Thibaut and Kelley's comparison level [37], the standard a relationship is judged against, is learned from experience, and Definition 11.1 has a ceiling, M_n, but no floor. Among 5,681 adolescents aged 12 to 18, dating violence was followed five years later by 1.9 to 3.6 times the adjusted odds of being a victim of partner violence, in both sexes, and psychological violence alone roughly doubled them [18]. This fits a lowered floor, but selection is not ruled out.

Remark 12.4 (Recovery). None of this need be permanent. In [15], attachment to the former partner fell by about 27% over the six months, so λ is better read as decreasing in the time since the relationship ended. An inability to experience loving feelings is a listed symptom of post-traumatic stress disorder [2], which needs no appeal to the used-up bonding capacity of Section 11. After breakups in general, not only abusive ones, sex to cope with distress or to get back at the ex is most frequent soon after the breakup and then declines [5].

Remark 12.5 (Flipping). A survivor is sometimes said to flip hypergamy and seek a partner of lower status. That is hypergamy for the status functional $-s$, so every theorem above applies with $-s$ in place of s, and by Theorem 2.2 the deficit becomes $\sup_t(\nu(s \le t) - \mu(s \le t))^+$. By Proposition 3.1, "he must be taller than me" is feasible for everyone, but in the calibration of Section 3, if all of $\mathfrak{W}$ requires "he must not be taller than me," about two thirds of $\mathfrak{W}$ must stay unmatched. Every union in which one partner marries down is one in which the other marries up. Whether survivors flip at all is not established. The nearest evidence concerns dominance rather than status, and depends on where the danger comes from: Colombian women who perceive more risk of domestic violence prefer less masculine Colombian faces [8], while American women more afraid of crime prefer more aggressive and formidable mates [35].

13 Conclusion

Hypergamy reduces to optimal transport on the real line. Its feasibility is stochastic dominance, its failure is the Hypergamy Deficit, its equilibrium is the quantile coupling, and its realised rate is the measure of the set where one quantile function exceeds another. Section 9 extends the deficit to several traits, where checklist requirements multiply when traits are independent and trade-offs reduce to a scalar index, and Section 10 reviews the evidence on the weights: stated preferences put kindness and intelligence first for the long run and looks first for the short run, money counts for whoever wants it, the relationship itself best predicts how it goes, and height enters as a relative comparison the market can satisfy. Section 11 treats partner history as a comparison effect that applies to both sexes, and Section 12 models an abusive ex as an idealised one, whose cost compounds where that of additional partners diminishes (Proposition 12.2). Open problems include general multidimensional status (Brenier maps and upper sets in a partial order), search frictions, and the case in which the status functional is itself endogenous.

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